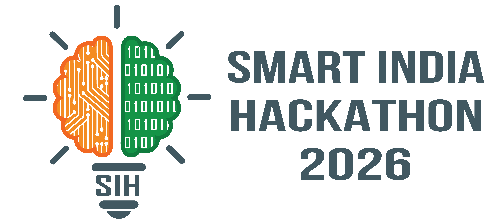


SMART INDIA HACKATHON 2026



SAMARATH

Coordinated railway maintenance planning

Problem Statement ID: SIH26027

**AI-Powered Automatic Block Planning to
Maximize Asset Availability for Train
Operations on Indian Railways**

Theme: Transportation & Logistics

PS Category: Software

Team Name: NIRVIKALP

Team ID: [ADD REGISTERED ID]





Proposed solution

Coordinated blocks across Engineering, Traction and Signalling

Separate maintenance requests compete for the same track time, crews and machines. SAMARATH proposes one reviewable programme.



How it addresses the problem

- Protects urgent work and train paths.
- Includes setup, testing and restoration.
- Checks crews, machines and readiness.
- Keeps officials in control of approval.

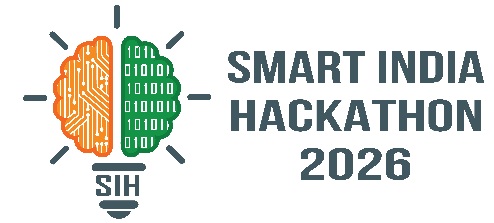
Innovation and uniqueness

- Explains exclusions and permitted repairs.
- Rechecks every proposal independently.
- Replans with fewer assignment changes.
- Unknown compatibility blocks grouping.

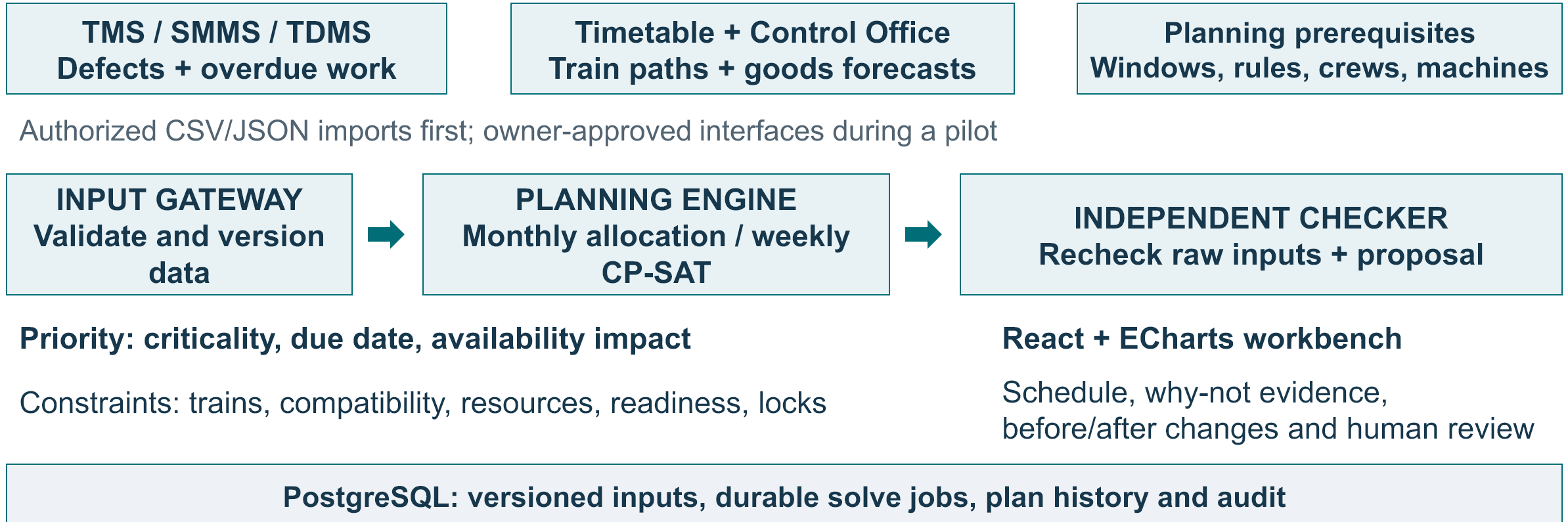
Why-not example: 75-minute job + 60-minute window = cannot fit. Suggest another approved window.



TECHNICAL APPROACH



System architecture and implementation workflow

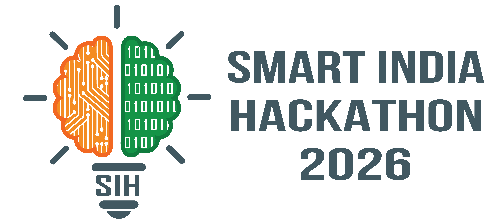


Changed input invalidates the old plan; a worker re-solves and rechecks before review.

Python / FastAPI + OR-Tools. Future ML: duration estimates from authorized history, tested against a baseline.



FEASIBILITY AND VIABILITY



Analysis of feasibility

A bounded pilot with existing software tools

Proposed TEST scope: 6 fictional stations, 3 departments, 60 monthly tasks, up to 30 per week.

Potential challenges and risks

Railway data and rule access

Conflicting work or unknown rules

Growing scheduling combinations

Changing trains or resources

Strategies for overcoming them

Start with labelled TEST data and approved exports.

Block unknown combinations. Independently check plans.

Limit the corridor, prune candidates, benchmark growth.

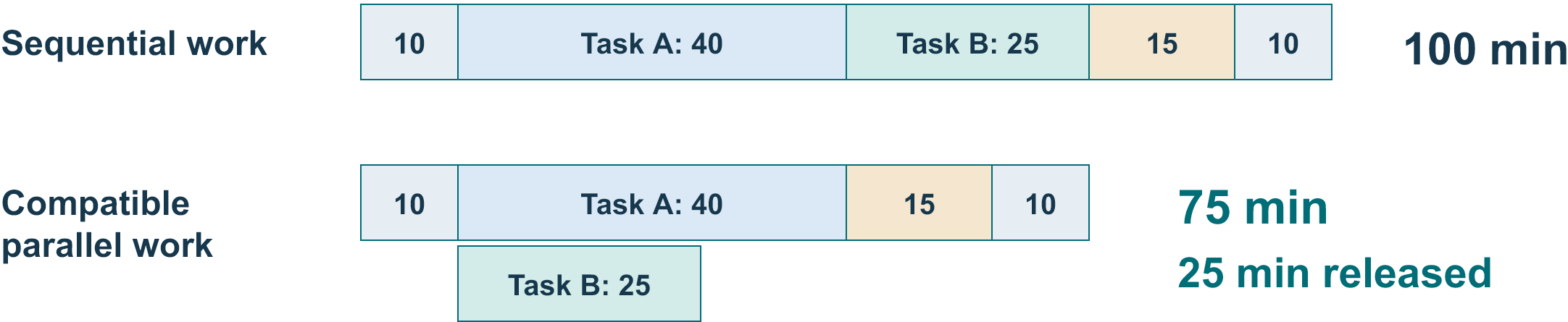
Mark old plans stale. Replan with fewer changes.

Pilot path: TEST validation, then authorized shadow planning, then measured expansion.

Potential impact on the target audience

More maintenance within the available block time

Illustrative example. Setup 10 min, testing 15 min, restoration 10 min in both cases.



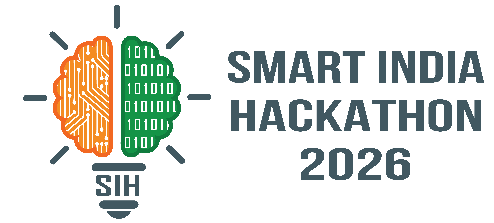
Only when compatibility, independent resources and common test/restoration assumptions hold.

Benefits: less occupation for operations, coordinated crews, and traceable review decisions.

Pilot measures: completed work, occupied minutes, missed deadlines and changed assignments. No field savings claimed.



RESEARCH AND REFERENCES



Research findings that shape the solution

Joint planning already exists. SAMARATH proposes better selection, explanations and revisions within that process.

01 Official SIH26027 requirements

Coordinated inputs, prioritization, weekly and monthly plans.

sih.gov.in/sih2026PS

02 Railway Board, 29 August 2023

Joint rolling blocks, resource preparation and official review.

iricen.gov.in/iricen/other_manual/ircm_ref_docs/1801.pdf

03 CRIS system descriptions

Existing maintenance and operating systems guide input design.

cris.org.in/loadpage?page=proTMS / [proTDMS](https://cris.org.in/loadpage?page=proTDMS) / [proCOA](https://cris.org.in/loadpage?page=proCOA)

04 Google OR-Tools CP-SAT documentation

Constraint scheduling with explicit feasible / optimal / unknown status.

developers.google.com/optimization/cp/cp_solver

Team research: 98-page design blueprint, rule/checker boundary, repair workflow and benchmark plan.

Prototype walkthrough: [ADD PUBLIC DEMO LINK]